

Mediation Analysis: MASTER CONSOLIDATED ANSWER KEY

Part 1: Foundations & The Conceptual Core

1. **application** (or implementation)
2. **mechanism** (or process)
3. **roadmaps**
4. **X** = Antecedent; **M** = Intermediary; **Y** = Consequent
5. **dependent** (predicted by X); **independent** (predicting Y)
6. **M**
7. **Paths:** a (X to M), b (M to Y), c' (X to Y)
8. **Scenario A: Positive (+) | Scenario B: Positive (+)**
 - **Math Explanation:** In A, $(+) * (+) = (+)$. In B, $(-) * (-) = (+)$.
 - **Conceptual Interpretation:** Both mechanisms show an overall increase in depression as food insecurity rises, even though the behaviors (visiting vs avoiding doctors) are opposite.

Part 2: Causal Assumptions & Logic

9. **X** → **M** → **Y**
10. **cross-sectional** (or correlational)

Part 3: The Statistical Model

11. **two**
12. **X; M**
13. **M; Y**
14. **direct**
15. **a; b**
16. **direct; indirect**

Part 4: Numeric Problem - Studying & Stress

17. **Indirect Effect:** **-6.0** (2.0×-3.0)
18. **Total Effect:** **-1.0** ($5.0 + (-6.0)$)
19. **Interpretation:** **Opposite direction** (The stress penalty outweighs the study benefit).

Part 5: Complete vs. Partial Mediation & Confounding

- 20. **disappears** (or becomes non-significant)
- 21. **directly; mediator**
- 22. **real**
- 23. **fake** (or spurious)
- 24. **Suppression**

Part 6: Testing & Bootstrapping

- 26. **total (c); indirect (ab)**
- 27. **Confidence**
- 28. **zero (0)**

Part 7: Case Study Exercises

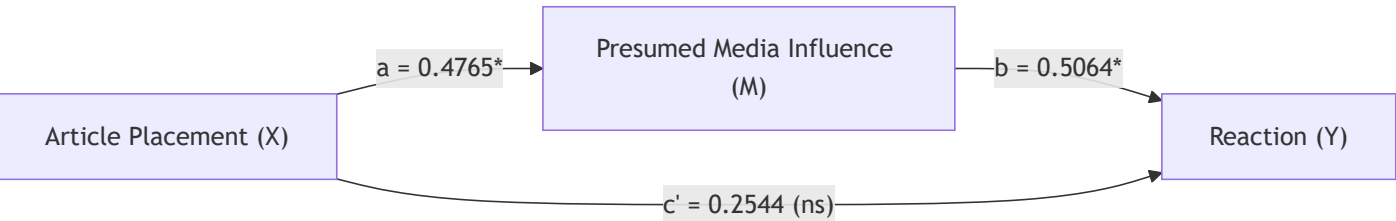
A. Example: Education & Income

- 29. **.5000**
- 30. **2000**
- 31. **500; .0480**
- 32. **.5000 * 2000 = 1000**
- 33. **1000**

B. The "Sugar Shortage" Study (PMI)

- 34. **.4765; .0454**
- 35. **.2544; .3221** (Non-significant)
- 36. **.5064; .0000** (Significant)
- 37. **Full** (or Complete) mediation (since c' is $p > .05$).
- 38. **Yes**; The interval **[.0095, .5304]** **does not contain zero**.

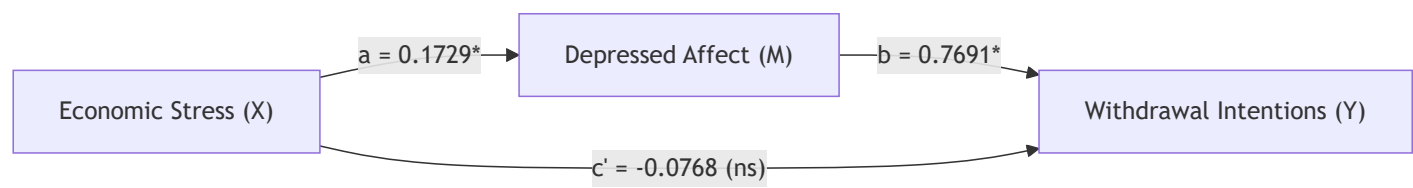
PMI Mediation Diagram:



C. Economic Stress (ESTRESS)

39. .1729 (a); .7691 (b)
40. .0561; .3015
41. **Yes** (Significant indirect effect is sufficient).
42. **Cancellation:** The positive indirect effect (+0.13) was negated by a negative direct effect (-0.08).

ESTRESS Mediation Diagram:



D. Legal Protest Study

1. **Hypothesis:** Protesting (X) increases perceived appropriateness (M), which increases evaluations (Y).
2. **Equations:**
 - $M = 4.043 + 0.798X$
 - $Y = 3.716 + (-0.118)X + 0.420M$

3. Summary of Model Coefficients:

Antecedent	Consequent: respappr(Mi)					Consequent: liking(Y)				
		Coef	SE	t	p		Coef	SE	t	p
protest(X)	a	0.798	0.128	6.248	<.001	c'	-0.118	0.113	-1.044	.298
respappr(Mi)						b	0.420	0.069	6.114	<.001
Constant	i	4.043	0.168	24.077	<.001	i	3.716	0.306	12.125	<.001

4. **Testing:**
 - **Indirect Effect:** 0.335 (0.798×0.420)
 - **Additive:** -0.118 (Direct) + 0.335 (Indirect) = 0.217 (Total)
5. **Interpretation:**
 - **R2:** 25.1% of the variance in Liking (Y) is explained by Protest and Appropriateness together.
 - **Support:** Yes. Significant positive indirect effect via appropriateness.

Part 8: Practical Software & Conventions

1. PROCESS
2. Model 4
3. Unstandardized

